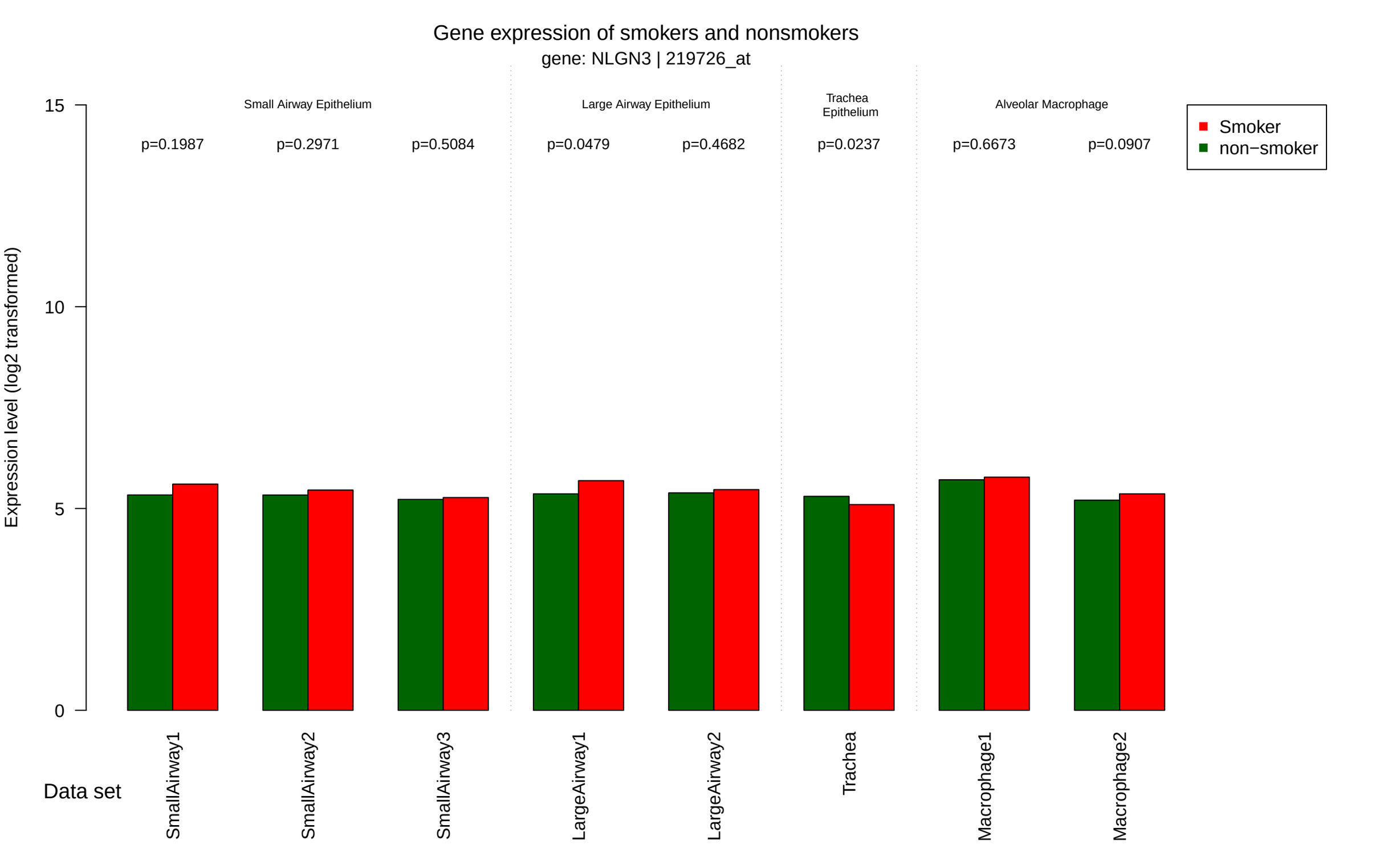
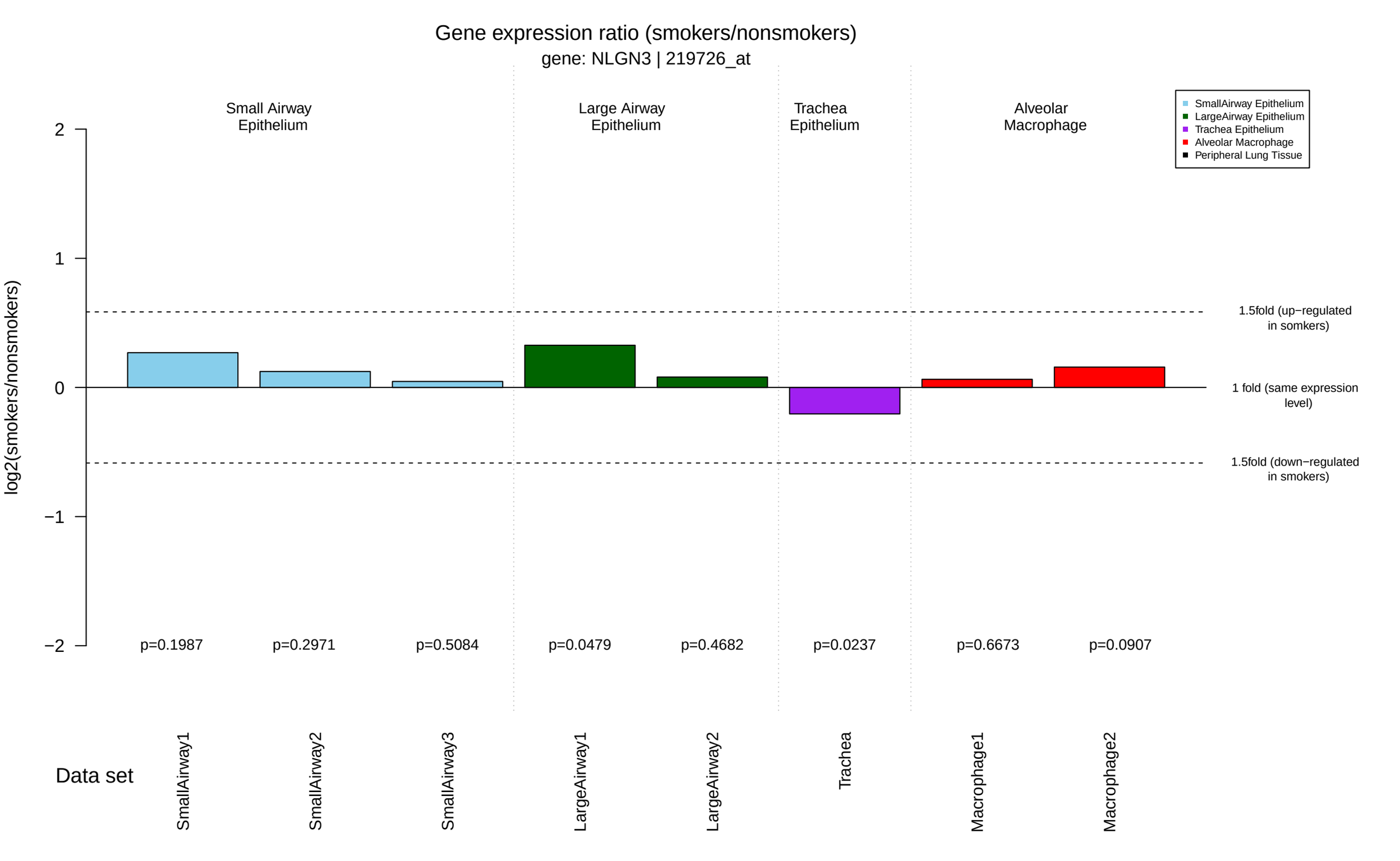
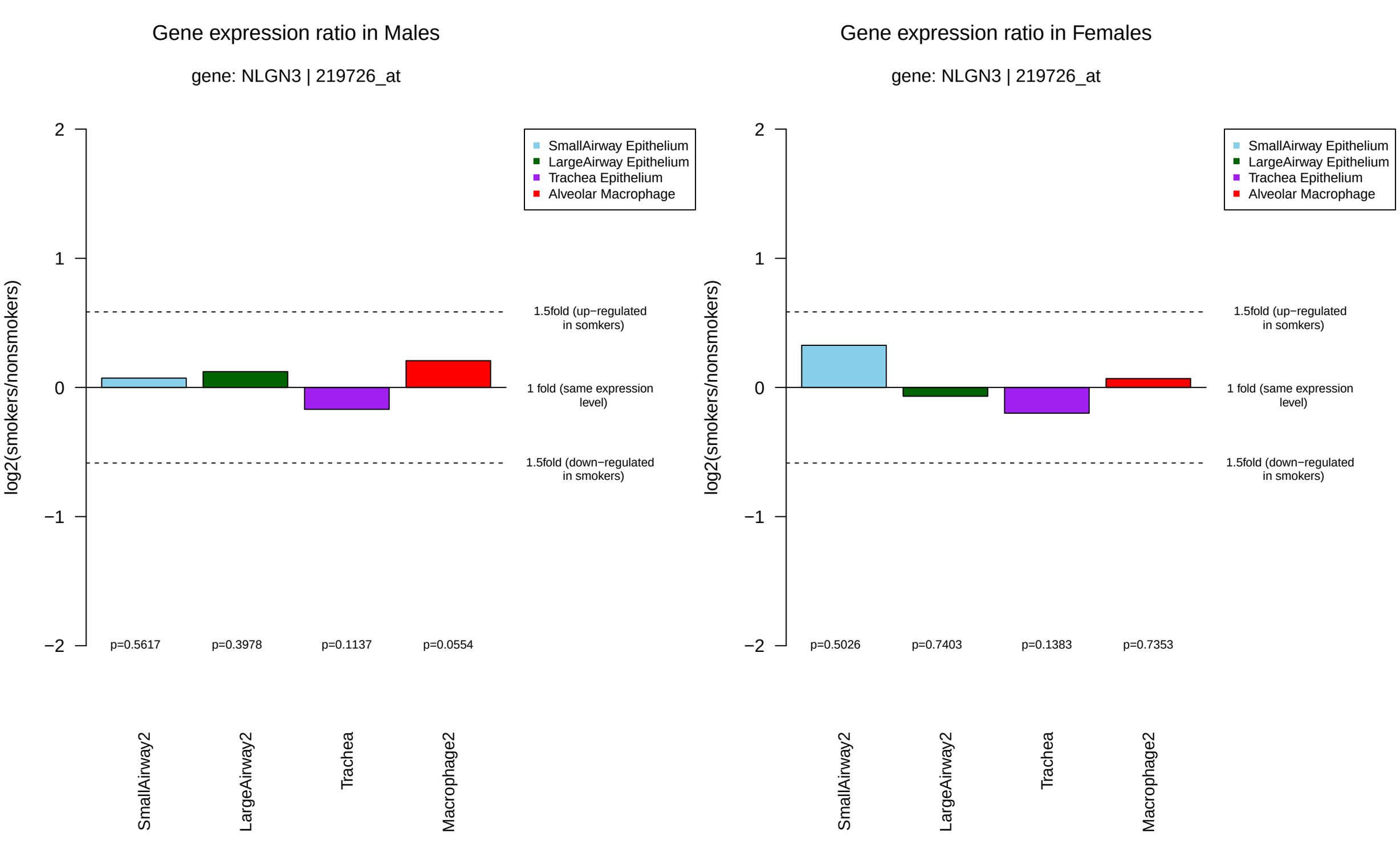
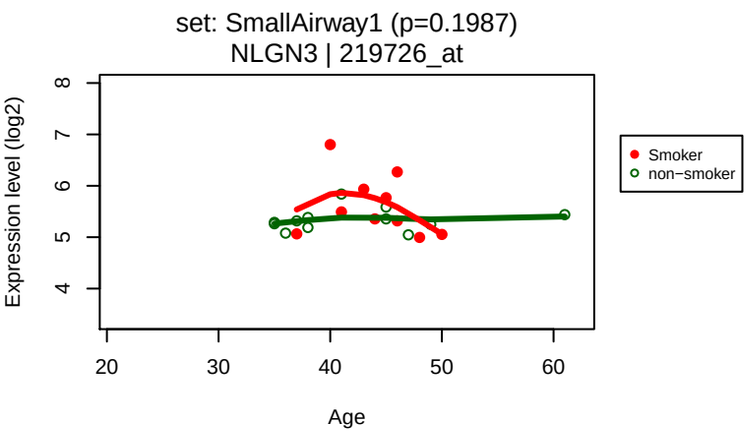
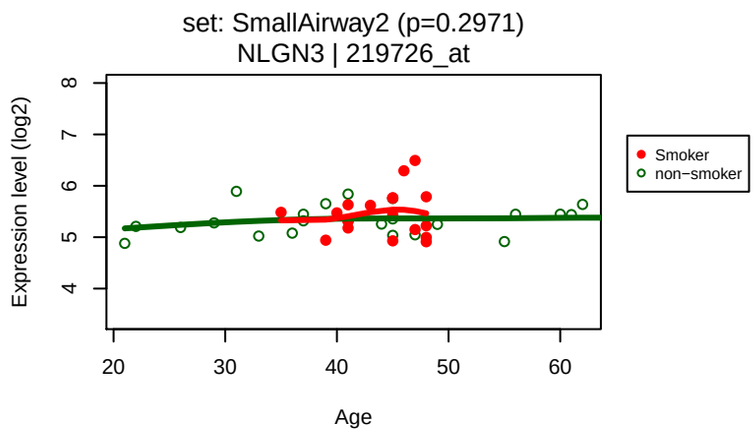
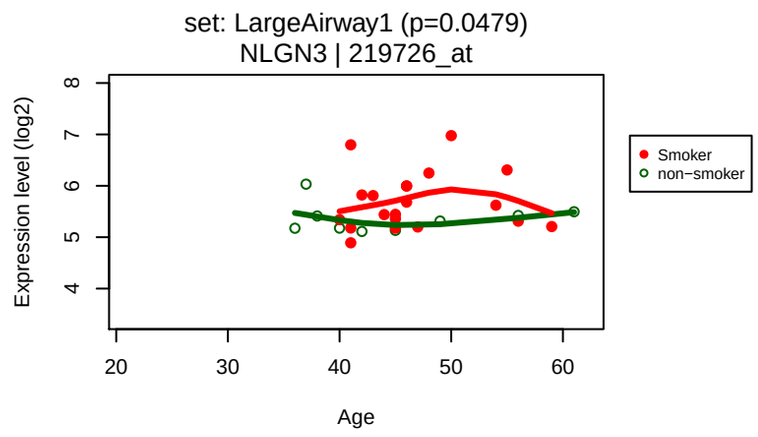

Gene expression of smokers and nonsmokers
gene: NLGN3 | 219726_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1987
p=0.2971
p=0.5084
p=0.0479
p=0.4682
p=0.0237
p=0.6673
p=0.0907
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

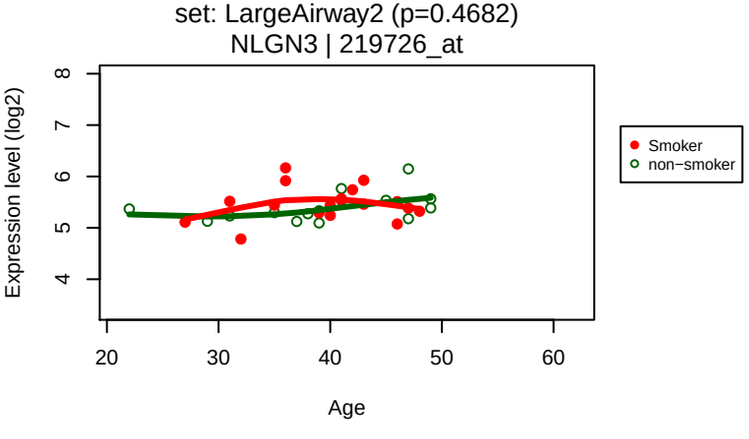

Gene expression ratio (smokers/nonsmokers)
gene: NLGN3 | 219726_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1987
p=0.2971
p=0.5084
p=0.0479
p=0.4682
p=0.0237
p=0.6673
p=0.0907
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

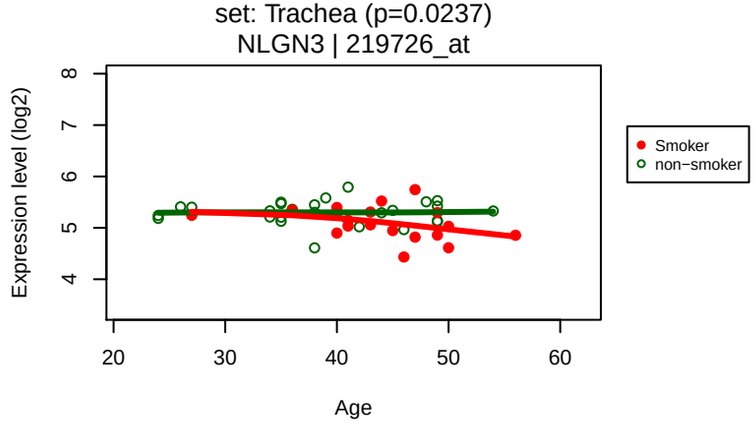

Gene expression ratio in Males
gene: NLGN3 | 219726_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5617
p=0.3978
p=0.1137
p=0.0554
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: NLGN3 | 219726_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5026
p=0.7403
p=0.1383
p=0.7353
SmallAirway2
LargeAirway2
Trachea
Macrophage2

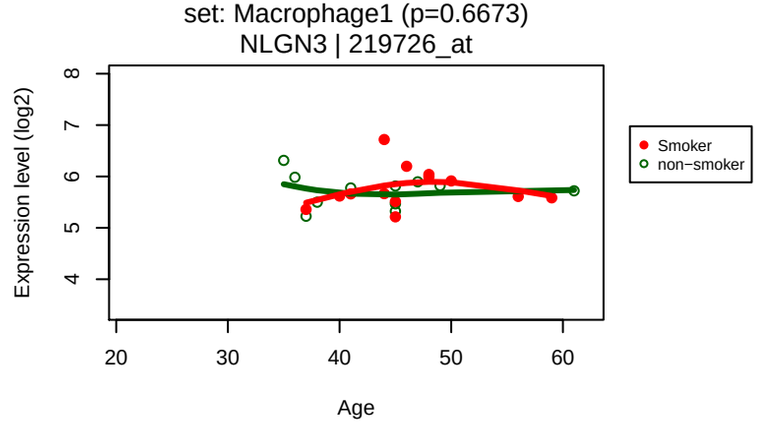

set: SmallAirway1 (p=0.1987)
NLGN3 | 219726_at
Expression level (log2)
Age
Smoker
non−smoker

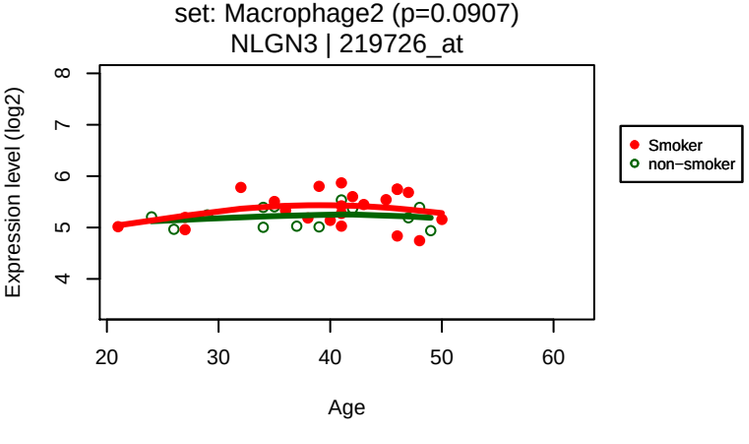

set: SmallAirway2 (p=0.2971)
NLGN3 | 219726_at
Expression level (log2)
Age
Smoker
non−smoker

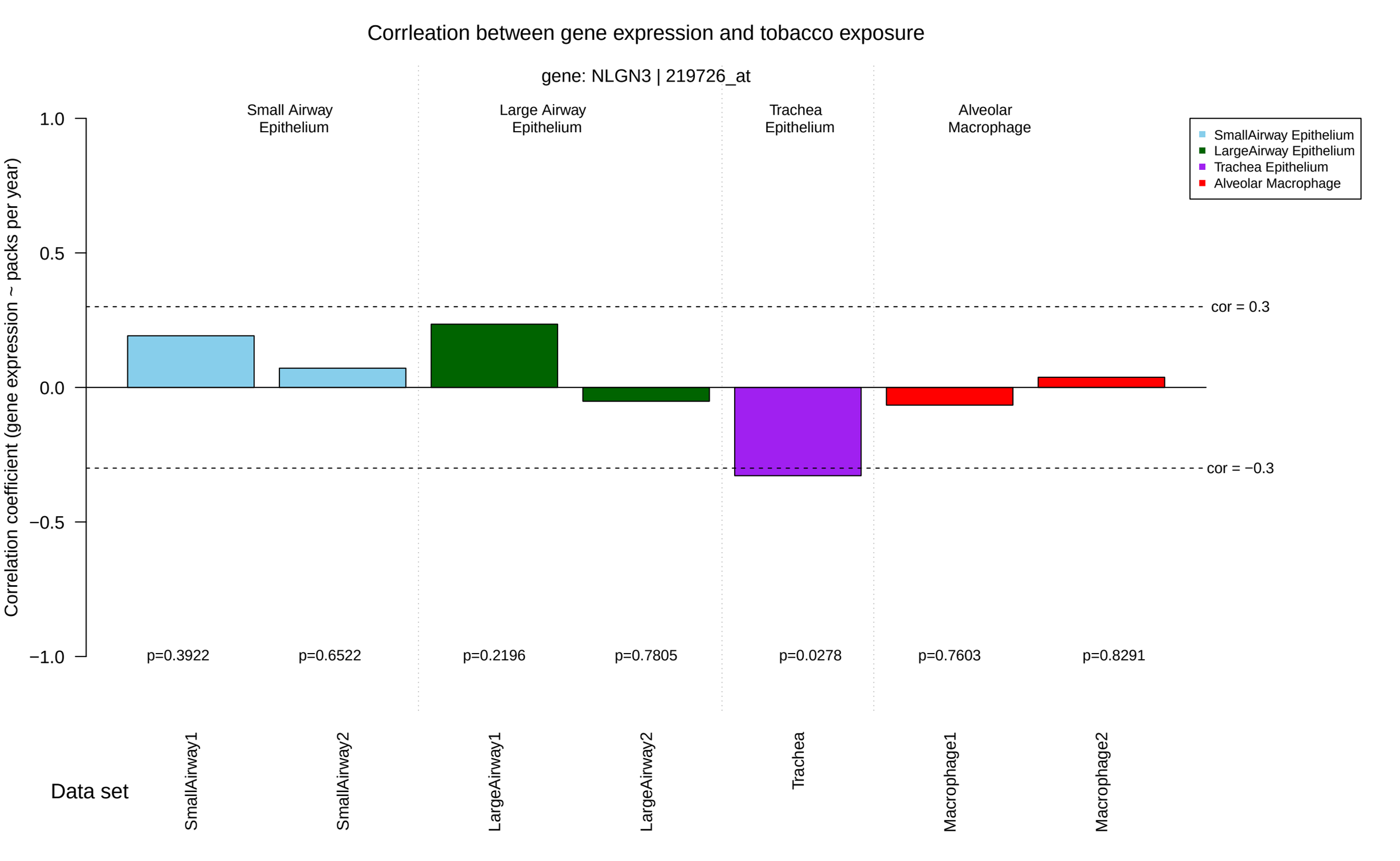

set: LargeAirway1 (p=0.0479)
NLGN3 | 219726_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4682)
NLGN3 | 219726_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0237)
NLGN3 | 219726_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6673)
NLGN3 | 219726_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0907)
NLGN3 | 219726_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: NLGN3 | 219726_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3922
p=0.6522
p=0.2196
p=0.7805
p=0.0278
p=0.7603
p=0.8291
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2